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Conclusion

Graph learning has emerged as a rapidly evolving and influential field within AI, offering powerful tools for modeling complex relationships and extracting valuable insights from both graph-structured and non-graph-structured data. This survey has provided a comprehensive review of state-of-the-art graph learning methods, covering key areas such as scalable graph learning, temporal graph learning, multimodal graph learning, generative graph learning, explainable graph learning, and responsible graph learning. Additionally, we have explored emerging topics, including graph foundation models, graph reinforcement learning, federated graph learning, learning on knowledge graphs, knowledge-infused graph learning, and quantum graph learning.

Recent research trends in graph learning, beyond the commonly discussed areas like scalability, temporal dynamics, multimodality, generative models, explainability, and responsibility, focus on several innovative directions with strong ties to AI advancements. Beyond the above-mentioned emerging topics, one key trend, for example, is causal graph learning and neurosymbolic integration. Causal graph learning seeks to go beyond mere pattern recognition by uncovering the underlying causal structures within graph data, which is crucial for robust

decision-making in AI systems. This direction aims to improve generalization, interpretability, and counterfactual reasoning capabilities, which are especially important in high-stakes domains such as healthcare, finance, and policy-making. Meanwhile, the fusion of graph learning with symbolic reasoning, often referred to as neurosymbolic graph learning, enables more structured, logic-aware AI by embedding domain knowledge and logical constraints into neural graph models, allowing for more trustworthy and knowledge-driven inference.

Another emerging trend is interactive and agent-centric graph learning, where the learning process is embedded within an interactive environment or driven by the actions of autonomous agents. This includes applications in reinforcement learning on graphs, active learning with graph feedback, and the use of graph-based world models for embodied AI or decision-making agents. Such models are increasingly relevant for robotics, scientific discovery, and multi-agent systems, where understanding the dynamics of relational structures is essential.

In the coming years, graph learning is poised to play a transformative role across a broad spectrum of real-world applications, driven by growing demands for, e.g., relational understanding, structured reasoning, and efficient learning. In domains such as healthcare, biology, and drug discovery, graph learning will enable more precise modeling of complex systems like molecular interactions and disease progression pathways, accelerating personalized medicine and therapeutic innovations. In transportation, urban computing, and supply chain management, graph-based models will become central to optimizing infrastructure, predicting disruptions, and enabling adaptive, intelligent planning.

Moreover, the integration of graph learning into large-scale systems such as recommender engines, fraud detection, cybersecurity, and knowledge-intensive search will expand as organizations seek AI solutions that can reason over rich, dynamic relational data. The increasing availability of heterogeneous and dynamic data sources will further fuel the adoption of graph-based AI in environmental monitoring, social behavior analysis, and scientific discovery. As these applications mature, future graph learning systems will likely become more autonomous, context-aware, and tightly coupled with decision-making processes, forming the backbone of next-generation intelligent systems.

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